MAT2002 Tutorial 7

Higher Order Linear Equations

Zha Junwei

CUHK(Shenzhen)

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Higher Order Linear Equation I

General nth order ODE:

$$y^{(n)} + P_{n-1}(t)y^{(n-1)} + \ldots + P_0(t)y = g(t)$$

with initial conditions:

$$y(t_0) = y_0,\ y'(t_0) = y_0',\ \ldots,\ y^{(n-1)}(t_0) = y_0^{(n-1)}$$

It is **homogeneous** if $g(t) = 0$, otherwise it is **non-homogeneous**.

Remark

The general theory for nth order linear equation is parallel to 2nd order case.

Definition (Wronskian)

Given any n functions $y_1(t), \ldots, y_n(t)$, define:

$$W(y_1, \ldots, y_n)(t) = \begin{vmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{vmatrix}$$

Remark

- *In general, $y_1, \ldots, y_n$ linearly independent does NOT imply $W \neq 0$, but the converse is always true.*
- *If $y_1, \ldots, y_n$ are solutions to homogeneous equation and $P_{n-1}(t), \ldots, P_0(t)$ are continuous on I, then $W \neq 0$ is equivalent to linear independence.*

Fundamental Set of Solutions I

Theorem

Consider the homogeneous equation

$$y^{(n)} + P_{n-1}(t)y^{(n-1)} + \ldots + P_0(t)y = 0$$

with initial conditions

$$y(t_0) = y_0,\ y'(t_0) = y_0',\ \ldots,\ y^{(n-1)}(t_0) = y_0^{(n-1)}$$

Suppose $P_{n-1}(t), \ldots, P_0(t)$ are continuous on I and $y_1(t), \ldots, y_n(t)$ are n solutions satisfying $W(y_1(t_0), \ldots, y_n(t_0)) \neq 0$, and $t_0 \in I$, then:

- *$y_1(t), \ldots, y_n(t)$ forms a fundamental set of solutions*
- *The general solutions are $y = c_1y_1 + c_2y_2 + \ldots + c_ny_n$*
- *The unique solution to the IVP can be determined by solving for c_i using the initial conditions*

Fundamental Set of Solutions II

Question

Can $W(y_1(t_0), \ldots, y_n(t_0)) \neq 0$ *for some* t_0 *and* $W(y_1(t_1), \ldots, y_n(t_1)) = 0$ *for some other* t_1*?*

Theorem (Abel's Theorem)

Suppose $y_1(t), \ldots, y_n(t)$ *are* n *solutions to the homogeneous equation*

$$y^{(n)} + P_{n-1}(t)y^{(n-1)} + \ldots + P_0(t)y = 0$$

then

$$W(y_1, \ldots, y_n)(t) = ce^{-\int P_{n-1}(t)dt}$$

for some $c \in \mathbb{R}$

Remark

The theorem means that the Wronskian is either always 0 or never 0.

Fundamental Set of Solutions III

Example

Verify that the given functions form a fundamental set of solutions:
(i) $y''' + 4y' = 0$ with $\{1, \cos(2t), \sin(2t)\}$

Sol: Check they are solutions,

$$W(1, \cos 2t, \sin 2t) = \begin{vmatrix} 1 & \cos 2t & \sin 2t \\ 0 & -2\sin 2t & 2\cos 2t \\ 0 & -4\cos 2t & -4\sin 2t \end{vmatrix}$$

$$= 8\sin^2 2t + 8\cos^2 2t = 8 \neq 0$$

Example

(ii) $y^{(4)} + 9y'' = 0$ with $\{1, t, \cos(3t), \sin(3t)\}$

Leave as exercise!

Structure for Solution Set (Non-homogeneous)

Theorem

The general solution of the non-homogeneous equation

$$y^{(n)} + P_{n-1}(t)y^{(n-1)} + \ldots + P_0(t)y = g(t)$$

is

$$y = c_1y_1 + c_2y_2 + \ldots + c_ny_n + Y(t)$$

where y_i's are fundamental set of solutions to homogeneous equation and $Y(t)$ is any particular solution.

Remark

This is similar to the theory of linear equation $AX = b$.

Homogeneous Equation with Constant Coefficients I

Consider

$$y^{(n)} + a_{n-1}y^{(n-1)} + \ldots + a_0 y = 0$$

By letting $y = e^{rt}$ we get the characteristic equation:

$$x^n + a_{n-1}x^{n-1} + \ldots + a_0 = 0$$

By fundamental theorem of algebra, the polynomial of degree n has n roots (real or complex, counted with multiplicity), denote them by $r_1, r_2, \ldots, r_n$

- If r is a real root with multiplicity m, then $e^{rt}, te^{rt}, \ldots, t^{m-1}e^{rt}$ are solutions
- If $r = \lambda + i\mu$ is a complex root with multiplicity m, then $\overline{r}$ is also a root with multiplicity m. Then:

$$e^{\lambda t}\cos(\mu t),\ te^{\lambda t}\cos(\mu t),\ \ldots,\ t^{m-1}e^{\lambda t}\cos(\mu t)$$

$$e^{\lambda t}\sin(\mu t),\ te^{\lambda t}\sin(\mu t),\ \ldots,\ t^{m-1}e^{\lambda t}\sin(\mu t)$$

are solutions

Examples

Example

Find the general solutions of

(i) $y^{(4)} - y = 0$

Sol: $r^4 - 1 = 0 \implies r = -i, i, 1, -1$ General solution:

$$y = c_1 \cos t + c_2 \sin t + c_3 e^{-t} + c_4 e^{t}$$

(ii) $y^{(4)} + 2y'' + y = 0$

Sol: $r^4 + 2r^2 + 1 = 0 \implies r = -i, -i, i, i$ (multiplicity 2) General solution:

$$y = c_1 \cos t + c_2 \sin t + c_3 t \cos t + c_4 t \sin t$$

Exercise

Exercise

Find the general solution of:

$$y^{(4)} + y^{(3)} - 7y'' - y' + 6y = 0$$

Solution: $r = -3, -1, 1, 2$

$$y(t) = c_1 e^{-3t} + c_2 e^{-t} + c_3 e^{t} + c_4 e^{2t}$$

Method of Undetermined Coefficients I

Consider

$$y^{(n)} + a_{n-1}y^{(n-1)} + \ldots + a_0 y = g(t)$$

Method of undetermined coefficients works for:

$$g(t) = P_m(t),\ e^{\alpha t},\ \cos(\beta t),\ \sin(\beta t)$$

Remark

This method is parallel to 2nd order case, the only difference is the multiplicity may be > 2.

$g(t)$	$Y(t)$	s
$P_m(t)$	$t^s Q_m(t)$	multiplicity of 0
$P_m(t)e^{\alpha t}$	$t^s Q_m(t)e^{\alpha t}$	multiplicity of α
$P_m(t)e^{\alpha t}\cos(\beta t)$ or $P_m(t)e^{\alpha t}\sin(\beta t)$	$t^s(Q_m(t)\cos\beta t + R_m(t)\sin\beta t)$	multiplicity of $\alpha + \mathrm{i}\beta$

Example: Undetermined Coefficients

Example

$y^{(4)} + 2y'' + y = 4\cos t - \sin t$

Sol: Characteristic polynomial: $r^4 + 2r^2 + 1 = 0 \implies r = -i, -i, i, i$
Corresponding to case 3, $\alpha + i\beta = i$ and $s = 2$ Let
$Y(t) = t^2(A\cos t + B\sin t)$
Substitute into the equation: $-8A\cos t - 8B\sin t = 4\cos t - \sin t$
$\implies A = -\frac{1}{2},\ B = \frac{1}{8}$
General solution:

$$y = c_1\cos t + c_2\sin t + c_3 t\cos t + c_4 t\sin t - \frac{1}{2}t^2\cos t + \frac{1}{8}t^2\sin t$$

Exercise

Write the form of a particular solution to:
(i) $y^{(3)} - y' = t^2 + 3\sin t + e^3 t$
(ii) $y^{(3)} + y^{(2)} + 4y' + 4y = 4t - 3te^{-t} + \cos 2t$

Solution

(i) $r = 0, 3, -3$ *and* $Y(t) = t\left(At^2 + Bt + C\right) + (D\cos t + E\sin t)t + tFe^{3t}$
(ii)$r = -1, 2i, -2i$ *and*
$Y(t) = At + B + t(Ct + D)e^{-t} + t(E\cos 2t + F\sin 2t)$

Method of Variation of Parameters I

Consider

$$y^{(n)} + P_{n-1}(t)y^{(n-1)} + \ldots + P_0(t)y = g(t)$$

We seek solution with the form:

$$Y(t) = u_1(t)y_1(t) + \ldots + u_n(t)y_n(t)$$

One can deduce from Cramer's rule that:

$$u'_m(t) = \frac{g(t)W_m(t)}{W(t)}$$

where $W(t)$ is the Wronskian, $W_m(t)$ is the determinant from $W(t)$ by replacing the m-th column by $(0, 0, \ldots, 1)^T$.

Method of Variation of Parameters II

Example

Find a particular solution of $y^{(3)} + y' = \sec^2 t$

Sol: $r^3 + r = 0 \implies r = 0, -i, i$

Fundamental set of solution for homogeneous equation is $\{1, \cos t, \sin t\}$

Suppose $Y(t) = u_1 \cdot 1 + u_2 \cos t + u_3 \sin t$

Then: $\begin{pmatrix} 1 & \cos t & \sin t \\ 0 & -\sin t & \cos t \\ 0 & -\cos t & -\sin t \end{pmatrix} \begin{pmatrix} u_1' \\ u_2' \\ u_3' \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \sec^2 t \end{pmatrix}$

Using Cramer's rule or Gauss-Jordan elimination:

$$\begin{pmatrix} u_1' \\ u_2' \\ u_3' \end{pmatrix} = \begin{pmatrix} \sec^2 t \\ \sec t \tan t \\ -\sec t \end{pmatrix} \implies \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} = \begin{pmatrix} \tan t \\ \ln|\sec t + \tan t| \\ -\sec t \end{pmatrix}$$

The particular solution is:

$$Y(t) = u_1 + u_2 \cos t + u_3 \sin t = -\cos t \cdot \ln|\sec t + \tan t|$$